AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-24 | Entry window 24 October 2026 (full day) | 10 records screened in scope

10 RECORDS IN SCOPE (67 EXCLUDED)

4 OF 10 SUBTOPIC BANDS POPULATED

2 EFFECT ESTIMATES WITH A CI

1 TIER-A RECORDS

3 RECORDS CARRIED ON METADATA ALONE

Signal of the day

65.2 million Medicare beneficiaries: the long-term $PM_{2.5}$–CVD slope is halved in ZIP-years that had wildfire smoke waves.

A national ZIP-year cohort (2017–2022) with 1 km total $PM_{2.5}$ and 10 km smoke $PM_{2.5}$ from machine-learning models; a smoke wave is $\geq$2 consecutive days with smoke $PM_{2.5}$ >10 $\mu g\,m^{-3}$. Raising the two-year mean from 5 to 9 $\mu g\,m^{-3}$: overall CVD first hospitalisation RR **1.09** (1.09–1.10) without smoke waves, **1.04** (1.03–1.04) with them; ratio of RRs 1.05 (1.04–1.06). The gap persists below 9 $\mu g\,m^{-3}$ and is largest in men, urban and western residents. **Where it is weak:** episodic spikes inflate a two-year mean without a proportional rise in risk, so the attenuation is partly a property of the exposure metric, not of smoke toxicity; ZIP assignment and a 10 km smoke grid blur the contrast further. **Why it matters here:** the long-term mean cannot represent a peaked exposure distribution. Peak-resolved, high-frequency network data are the input this kind of model lacks.

What changed today

- **Wildfire PM toxicology is mostly not wildfire PM.** Systematic review of 46 studies (2005–2025): only 42 % of in-vitro and 27 % of in-vivo studies used particles collected during real fires; no chronic or toxicokinetic work and no mixtures with urban PM.
- **Brown carbon toxicity tracks lipophilicity.** 350 °C wood-pyrolysis BrC, 776–1,132 molecular features per extract: A549 cytotoxicity rose water < methanol < acetone; methoxyphenols, quinols and coniferylaldehyde named as contributors.
- **Cordwood stove trade-off (metadata only).** Automated forced-air injection lowers $PM_{2.5}$ mass but shifts emissions to smaller particles — the regime where optical mass sensors overstate the gain.
- **Five cross-day duplicates moved to 23 October.** Records harvested again in the 24 October window (EPMC re-surfacings of 23 October PubMed entries) were assigned to the earlier issue; one smoke–dementia preprint had already shipped on 12 October and was blocked by the DOI gate.
- **Saturday floor.** PubMed returned 2 records; Europe PMC (40) and the Crossref ISSN sweep (33) carried a thin window, and 3 of 10 admitted records have no abstract anywhere.

Window continuity

The 23 October issue closes the window through 23 October. This issue collects **24 October in full**, continuous with it. It was built on the **30 October 2026 recovery run** (14:20 UTC (09:20 CDT)), which backfilled 23–29 October day by day after the scheduled runs of 24–29 October did not complete; a 30 October 00:21 UTC run had harvested 23–28 October but stopped before screening. 24 October is a Saturday: the **W43 weekly (18–24 October)** is filed under this date as a separate report.

Corpus analytics

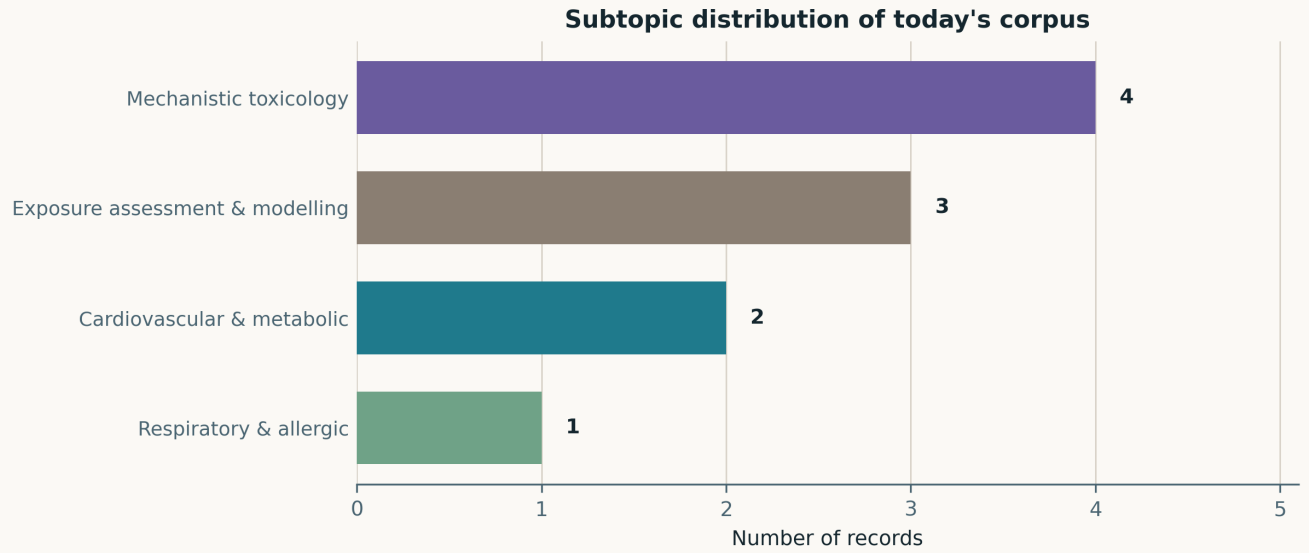


Fig. Subtopic assignment of the 10 in-scope records across four bands. Mechanistic toxicology leads with four (wildfire-PM review, brown-carbon cytotoxicity, gut microbiota, rat lung); exposure assessment three (all metadata-only); cardiovascular two; respiratory one. Sensing is empty today.

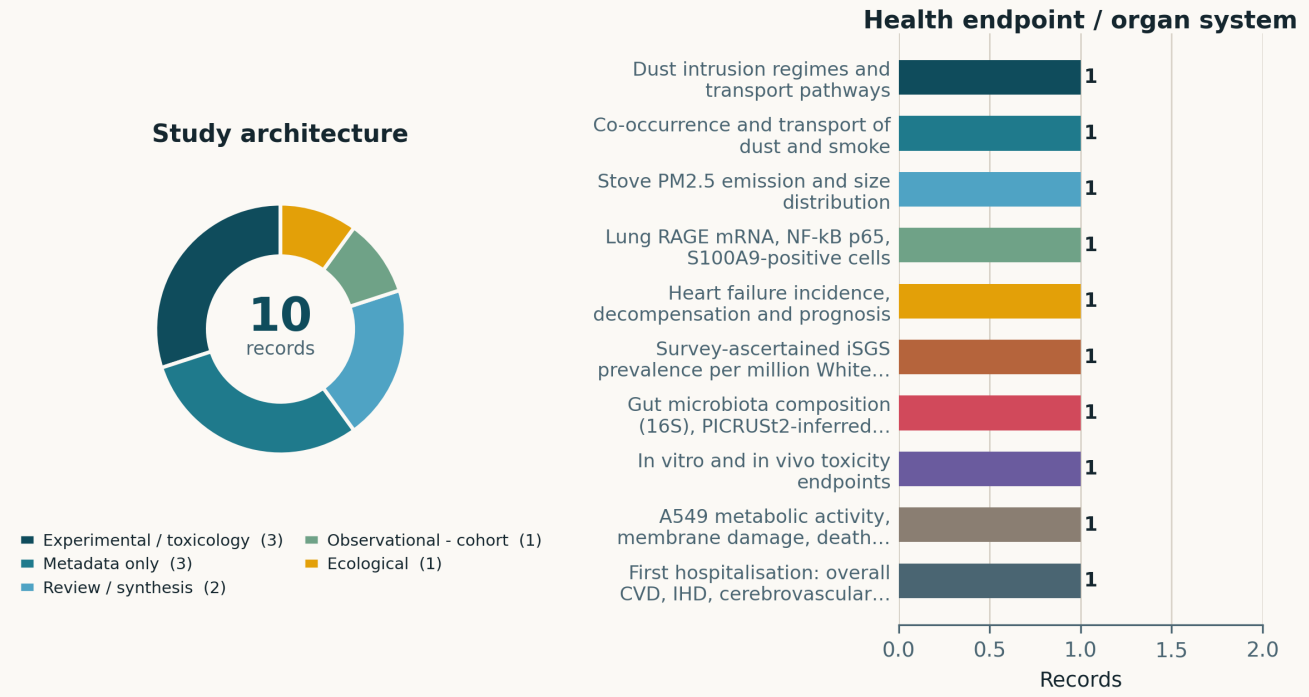


Fig. Left — architecture: two animal studies, one in-vitro, one systematic review, one narrative review, one nationwide cohort, one ecological analysis; **metadata-only 3 (30 %)**. Right — endpoints: CVD hospitalisation, heart failure, subglottic stenosis prevalence and in-vitro/in-vivo toxicity markers.

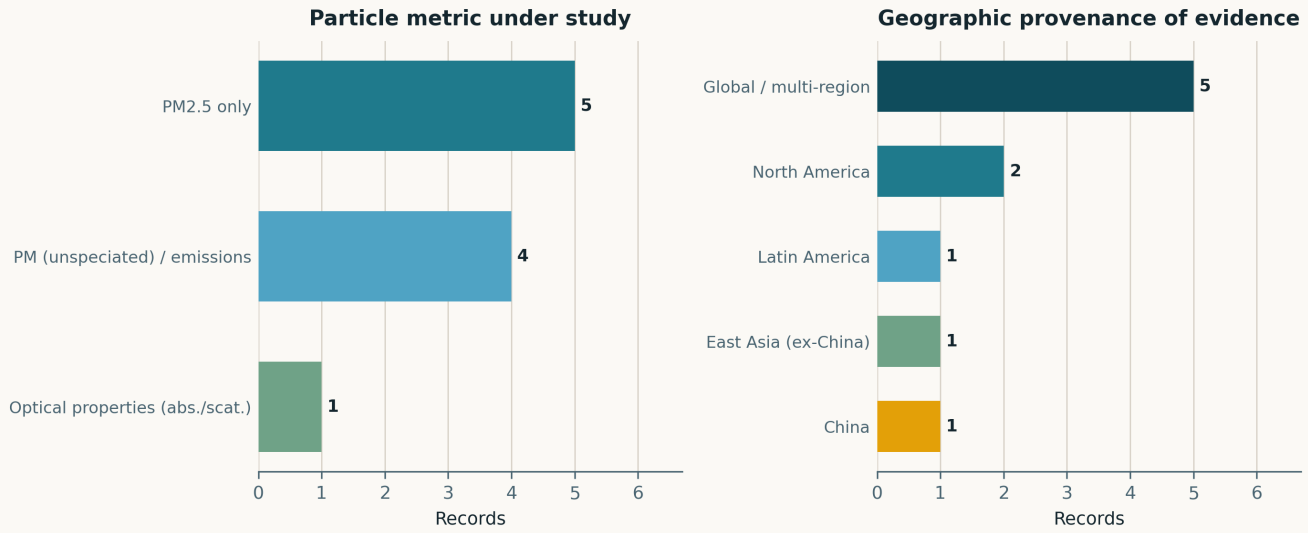


Fig. Left — particle metric: $PM_{2.5}$-only dominates; dust/smoke aerosol and stove emissions are unspeciated. Right — geography: North America 2 (Medicare; iSGS), global 2 (reviews), East Asia, Argentina and China one each; the rest are laboratory or not stated.

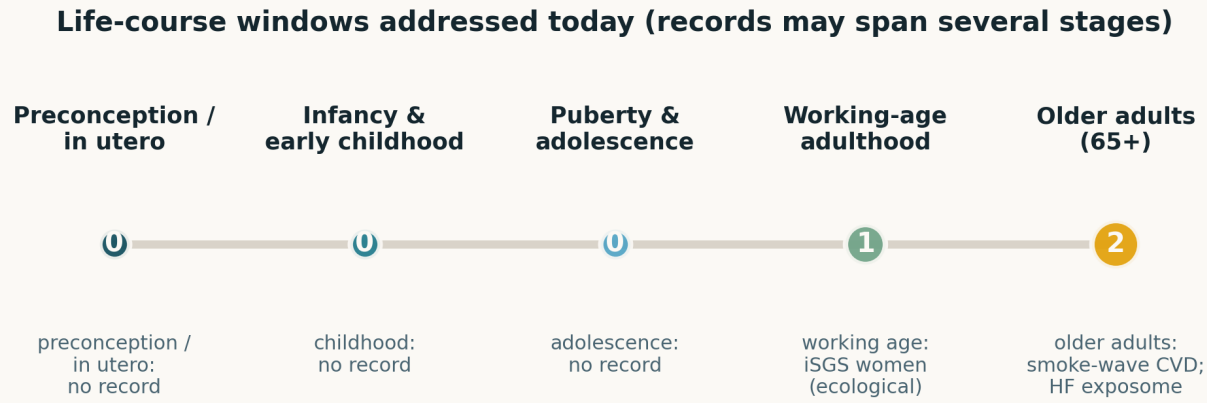


Fig. Life-course windows: two older-adult records (smoke-wave CVD; heart-failure exposome) and one working-age ecological record (iSGS in women). No in-utero, childhood or adolescent record today.

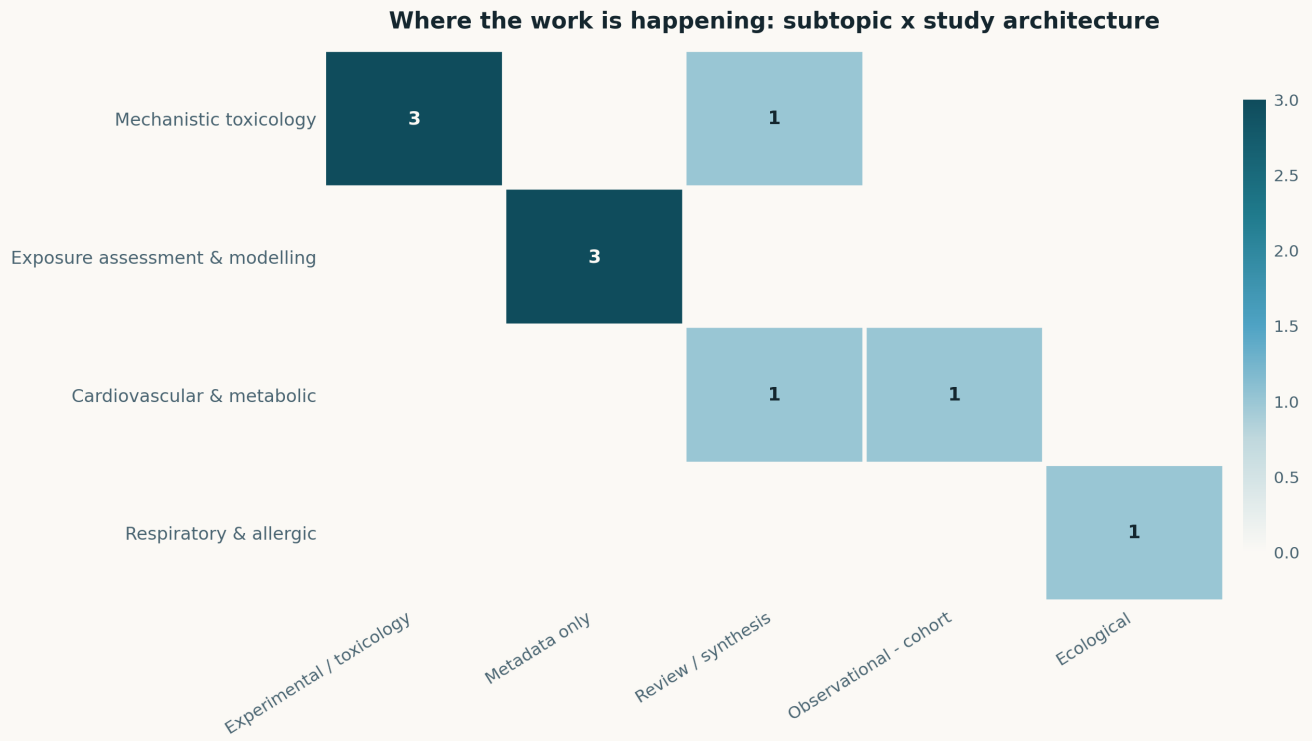


Fig. Subtopic × study architecture. Toxicology sits in animal, in-vitro and review cells; the two cardiovascular records split between a cohort and a narrative review; the exposure band is entirely metadata-only.

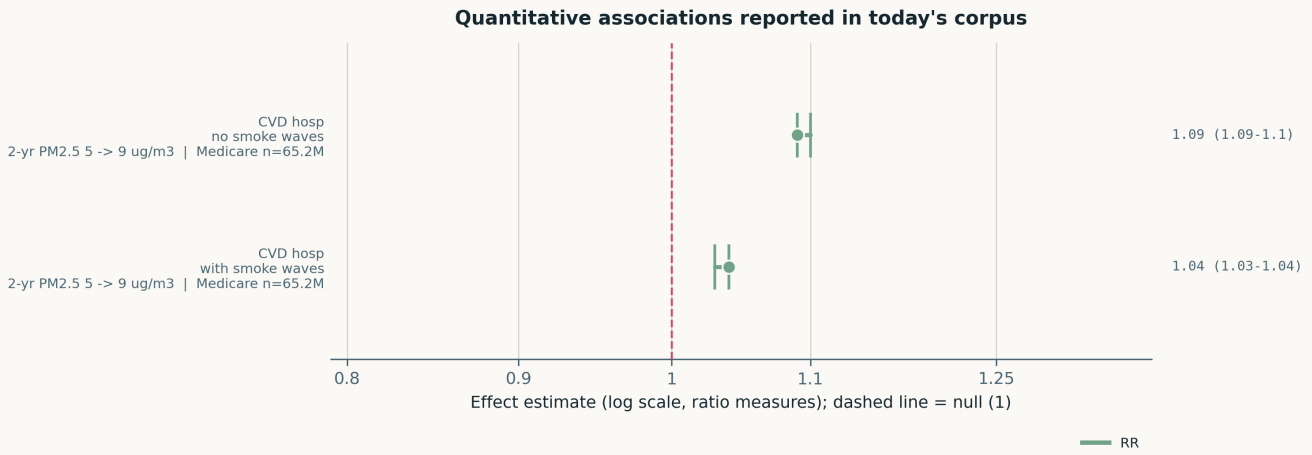


Fig. Ratio scale, two estimates from one study: overall CVD first-hospitalisation RR for a two-year $PM_{2.5}$ increase from 5 to 9 $\mu g\,m^{-3}$ in Medicare ZIP-years without and with wildfire smoke waves. Same cohort, stratified — the contrast, not either estimate, is the finding.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Cardiovascular & metabolic — 2 records

Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare 2026 Environ Res Health

Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States

65.2 million Medicare beneficiaries aged >=65, 2017-2022, ZIP-year quasi-Poisson. Raising two-year $PM_{2.5}$ from 5 to 9 $\mu g\,m^{-3}$ (WHO guideline to US standard) gave overall CVD RR 1.09 (1.09-1.10) in ZIP-years without smoke waves but 1.04 (1.03-1.04) with them (ratio of RRs 1.05, 1.04-1.06); the pattern held below 9 $\mu g\,m^{-3}$ and was stronger in men, urban and western residents. Limitation: the attenuation is what one expects if smoke spikes inflate the two-year mean with a less toxic-per-mass or differently distributed exposure, so this is an exposure-metric artefact as much as a biological finding; ZIP-level assignment and 10 km smoke grids blur the contrast. Relevance: high - long-term means hide episodic peaks that only high-frequency network data resolve.

doi: 10.1088/2752-5309/aeb0c3

Hahad exposome heart failure older adults review 2026 Curr Heart Fail Rep

Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives

Narrative review arguing for a life-course, systems-level view of heart failure in older adults in which air pollution sits alongside traffic noise, temperature extremes, metals, artificial light at night and social environment. Mechanisms proposed to converge on oxidative stress, mitochondrial dysfunction, inflammaging, impaired autophagy, epigenetic change and senescence, lowering the threshold for HF onset and decompensation. No new data or pooled estimates. Limitation: the mechanistic convergence is asserted from separate literatures; co-exposures (PM, noise, heat) are spatially correlated, and the review does not address how to separate them. Relevance: low - multi-exposure co-located sensing (PM, noise, temperature) is what the exposome framing would need.

doi: 10.1007/s11897-026-00781-w

Respiratory & allergic — 1 record

Harb idiopathic subglottic stenosis geographic clustering ecological 2026 Laryngoscope Investig Otolaryngol

Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study

4,450 members of an online iSGS support group (99.3% female) mapped to state/province and scaled by census White female populations. Prevalence clustered in dry, high, warm areas: Wyoming 18.1, Utah 17.4, Colorado 13.9 per million vs New Hampshire 1.2; Alberta 85.4 vs Quebec 2.9. $PM_{2.5}$ was not independently associated with prevalence but modified the elevation association. Limitation: the numerator is self-selected membership of an online group, so 'prevalence' reflects diagnosis, awareness and referral patterns; state-level ecology cannot separate climate from health-system access. Level 4 evidence by the authors' own grading. Relevance: low - a null for PM at state scale, hypothesis-generating only.

doi: 10.1002/lio2.70607

Mechanistic toxicology 4 records

Zendi brown carbon wood pyrolysis tracers cytotoxicity A549 2026 Chem Res Toxicol

Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells

Smouldering-regime BrC was generated by 350 C wood pyrolysis, extracted in three solvents and characterised by UHPLC-ESI-ToF-MS non-target analysis (776-1,132 molecular features per extract). Water extracts were dominated by lignin phenols and aromatic acids; methanol and acetone pulled less oxidised terpenoid and sterol-like species. Cytotoxicity in A549 cells ranked water < methanol < acetone, consistent with lipophilic, membrane-partitioning and redox-active compounds; named contributors to baseline narcosis included 3-hydroxybenzoic, salicylic and p-coumaric acids, benzene-1,2,4-triol and coniferylaldehyde. Limitation: solvent extracts at unreported dose-equivalence to inhaled mass, a single carcinoma line, and pyrolysis rather than flaming or aged smoke. Relevance: low-medium - supports BrC (UV-absorbing) channels as a toxicity-weighted metric.

doi: 10.1021/acs.chemrestox.6c00273

de Luna wildfire PM toxicity systematic review 2026 J Hazard Mater

Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward

46 in vitro and in vivo studies, 2005-2025 (epidemiology excluded). Only 42% of in vitro and 27% of in vivo studies used PM actually collected during wildfires; the rest used surrogate burns. In vitro work concentrates on lung (50%) and macrophages (17%); inhalation studies on lung (46%), heart (15%) and brain (15%). Mechanisms reported are mainly inflammation, oxidative stress and DNA damage, and toxicity varied with dose, size, fuel and burn condition. Gaps: incomplete physicochemical characterisation, unrealistic doses, few chronic or toxicokinetic studies, no work on mixtures with urban pollution. Limitation: no quantitative synthesis or risk-of-bias grading, so heterogeneity is described, not explained. Relevance: medium - the field cannot yet say which smoke property a sensor should track beyond mass.

doi: 10.1016/j.jhazmat.2026.143985

Ruan gut microbiota dose-time trajectories $PM_{2.5}$ PFOS mouse 2026 Environ Pollut

Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in $PM_{2.5}$ and PFOS exposure models

Two mouse models ($PM_{2.5}$ inhalation; PFOS in drinking water) sampled repeatedly across dose levels and exposure stages, with 16S sequencing and generalised additive models for dose-time structure. Controls showed clear temporal succession, which the authors stress must be modelled before attributing change to exposure. In the $PM_{2.5}$ model, restructuring peaked at early-to-intermediate stages, with nonlinear trajectories in Lactobacillus, Prevotella and the Firmicutes/Bacteroidetes ratio; PFOS effects were later and longer. No effect sizes with CIs are reported in the abstract. Limitation: functional pathways are inferred, not measured; doses and particle source are not given in the abstract, and cage effects dominate small murine microbiome studies. Relevance: low - mechanistic, but the time-varying response argues against single-timepoint toxicology.

doi: 10.1016/j.envpol.2026.129383

Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats 2026 Pak J Pharm Sci

Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats

40 male Wistar rats in five groups of 8. $PM_{2.5}$ at 500 $\mu g\,m^{-3}$ raised lung RAGE mRNA 2.8-fold (warm-dry) and 2.5-fold (cool-dry), NF-kB p65 2.4- and 2.2-fold, and S100A9-positive cells to 3.45% and 3.13% vs 0.82% in controls (all $P<0.05$). The two herbal formulas, given by gavage, partly reversed these markers in their matched climate model (e.g. RAGE 1.6-fold with Sang Xing Tang). Limitation: 500 $\mu g\,m^{-3}$ is an episode-level dose, n=8 per group, no histology scoring or blinding reported, and formula composition is not standardised. Relevance: low - a temperature-humidity x PM interaction in toxicology, not monitoring.

doi: 10.36721/pjps/2026.40.15845.1

Exposure assessment & modelling 3 records

de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size 2026 Atmos Environ

Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove

METADATA ONLY. Metadata only: no abstract was available from PubMed, Europe PMC, Crossref or Semantic Scholar at harvest. The title reports that automated forced-air injection in a residential cordwood stove lowers $PM_{2.5}$ mass emissions but shifts particles to smaller sizes. No numbers, test protocol or fuel details can be summarised. Limitation: cannot be appraised from the title; the trade-off claim needs number-based metrics to evaluate. Relevance: high in principle - a mass reduction with a shift to ultrafine sizes is precisely the case where optical mass sensors, blind below ~0.3 um, would overstate the improvement.

doi: 10.1016/j.atmosenv.2026.122443

Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific 2026 Atmos Environ

Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific

METADATA ONLY. Metadata only: no abstract was available at harvest from any route tried. The title describes convergence of mineral dust and biomass-burning aerosol over East Asia and the western North Pacific, presumably from observation and transport analysis; the instruments, period and metrics cannot be stated. Limitation: not appraisable. Relevance: medium in principle - mixed dust-smoke plumes combine coarse non-spherical and fine absorbing particles, a regime where optical sensor size and refractive-index assumptions fail together.

doi: 10.1016/j.atmosenv.2026.122444

Mulena dust intrusion regimes north-central Argentina vertical structure optical properties 2026 Atmos Res

Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways

METADATA ONLY. Metadata only: no abstract at harvest. The title describes a classification of dust-intrusion regimes over north-central Argentina by vertical structure, optical properties and transport pathway, likely from lidar or sun-photometer records with trajectory analysis. No data can be summarised. Limitation: not appraisable; column and vertical optical properties do not by themselves give surface PM. Relevance: low-medium - dust-regime classification is the prior a satellite-surface fusion model needs for coarse-mode episodes in an under-monitored region.

doi: 10.1016/j.atmosres.2026.109404

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Hahad exposome heart failure older adults review	Exposomic Determinants in Heart Failure Among Older Adults - Mechanistic Links, Clinical Evidence, and Future Perspectives	*Curr Heart Fail Rep*	Narrative review	Air pollution (PM) within a multi-exposure exposome (noise, heat, metals, light at night, greenness)	Global / multi-region
Zhang wildfire smoke waves long-term $PM_{2.5}$ cardiovascular hospitalisation Medicare	Impact of wildfire smoke waves on the association between long-term exposure to fine particulate matter and cardiovascular hospitalizations among older adults in the United States	*Environ Res Health*	Nationwide cohort	Two-year mean $PM_{2.5}$ (1 km ML model) and smoke $PM_{2.5}$ (10 km); smoke wave = >=2 d with smoke $PM_{2.5}$ >10 $\mu g/m^3$	USA (contiguous)
Respiratory & allergic					
Harb idiopathic subglottic stenosis geographic clustering ecological	Geographic Clustering of Idiopathic Subglottic Stenosis in North America: An Ecological Cohort Study	*Laryngoscope Investig Otolaryngol*	Ecological	State/province $PM_{2.5}$ alongside temperature, humidity and elevation	North America (USA and Canada)
Mechanistic toxicology					
Kou Xing Su San Sang Xing Tang $PM_{2.5}$ lung injury rats	Xing Su San and Sang Xing Tang attenuate $PM_{2.5}$-induced lung injury and inflammatory marker expression in rats	*Pak J Pharm Sci*	Animal exposure study	$PM_{2.5}$ inhalation 500 $\mu g/m^3$, 3 h/day, 28 days, under warm-dry or cool-dry conditions	China (laboratory)
Ruan gut microbiota dose-time trajectories $PM_{2.5}$ PFOS mouse	Time-resolved profiling reveals nonlinear dose-time trajectories of gut microbiota in $PM_{2.5}$ and PFOS exposure models	*Environ Pollut*	Animal exposure study	$PM_{2.5}$ inhalation at multiple dose levels (separate PFOS drinking-water model)	Not stated
Zendi brown carbon wood pyrolysis tracers cytotoxicity A549	Fine Brown Carbon Aerosol Emitted by Wood Pyrolysis - New Harmful Tracers and Cytotoxicity in Human Lung Cells	*Chem Res Toxicol*	In vitro (cell lines)	Fine brown carbon from 350 C wood pyrolysis; water, methanol and acetone extracts	Not applicable (laboratory)
de Luna wildfire PM toxicity systematic review	Systematic review on the toxicity of wildfire emitted particulate matter: State of science, knowledge gaps and way forward	*J Hazard Mater*	Systematic review (PRISMA)	Wildfire-emitted PM (field-collected or laboratory surrogate)	Global / multi-region
Exposure assessment & modelling					
Mulena dust intrusion regimes north-central Argentina vertical structure optical properties	Dust intrusion regimes over north-central Argentina: Vertical structure, optical properties, and transport pathways	*Atmos Res*	Metadata only (no abstract)	Mineral dust aerosol vertical profiles and optical properties	Argentina (north-central)
Tsai mineral dust biomass burning aerosol convergence East Asia western North Pacific	Convergence of mineral dust and biomass-burning aerosols over East Asia and the western North Pacific	*Atmos Environ*	Metadata only (no abstract)	Mixed mineral dust and biomass-burning aerosol (transport)	East Asia
de la Cruz automated forced air injection cordwood stove $PM_{2.5}$ particle size	Less Mass of $PM_{2.5}$ Emissions, Smaller Particles: The Trade-Off of Automated Forced Air Injection in a Residential Cordwood Stove	*Atmos Environ*	Metadata only (no abstract)	$PM_{2.5}$ mass emission and particle size distribution	Not stated

Provenance and method

Entry window **24 October 2026**, single day (Saturday). Harvester legs: **PubMed** health **2**, sensing **0**; **Europe PMC 40** (4 AGRICOLA retro-index dropped); **Crossref by ISSN 33**; **arXiv** 150 kB, no in-window entry; **OpenAlex** HTTP 429. PubMed connector (EDAT 24 Oct) 11 PMIDs, 8 appended; **77 screened**. Abstracts for 4 records recovered via Crossref/Semantic Scholar title match. **10 admitted**, **67 rejected** with logged reasons, including 5 cross-day duplicates assigned to 23 October and 1 DUPLICATE-DOI (shipped 12 October). **DOI gate** 0 fail / 0 warn after removal. **3 carried on metadata alone**. Consensus sweep and trial watch for the 23–29 October backfill are reported in the 29 October issue and the W43 weekly. **Scholar Gateway** not called. Abstracts remain the copyright of their publishers.